AI-BASED EARLY MENTAL HEALTH BREAKDOWN DETECTION FROM SPEECH PATTERNS

Data Science and AI Lab Project
(BSDA4001)
Milestone 2 – Dataset Preparation

INTRODUCTION

Early detection of mental health deterioration is difficult using traditional screening methods.

Current methods rely on:

- Self-reported questionnaires (PHQ-8)
- Subjective responses
- Recall bias and social desirability bias

Goal of this project

- Detect early mental health breakdown using speech and conversational patterns
- Provide an objective AI-based screening tool

PROJECT OBJECTIVE

The system aims to:

- Detect early signs of depression
- Analyze speech + conversational content
- Combine audio and language signals
- Assist clinicians with non-invasive screening

Key idea:
Use multimodal AI models to analyze behavioral markers in speech.

SYSTEM ARCHITECTURE OVERVIEW

The proposed system consists of three major components:

1. Speech Analysis Pipeline
2. Transcript-Based Analysis
3. RAG-Based Reasoning System

These modules are combined using Gated Dual-Modal Fusion.

SPEECH ANALYSIS PIPELINE

This module processes raw audio recordings from clinical interviews.

Extracted acoustic features include:

- Speech rate
- Pause patterns
- Vocal energy
- Prosodic variations
- Speech rhythm changes

Model used:
OpenAI Whisper Encoder

Purpose:
Extract audio embeddings capturing depression-related vocal biomarkers.

TRANSCRIPT-BASED ANALYSIS

Audio interviews are accompanied by time-aligned transcripts.

The transcripts help detect:

- Negative self-referential language
- Expressions of sadness or hopelessness
- Cognitive patterns linked to depression
- Changes in emotional tone across conversations

Model used:
BioClinical-ModernBERT

Advantages:

- Handles long conversations
- Captures contextual meaning

RAG-BASED REASONING SYSTEM

A Retrieval-Augmented Generation (RAG) module provides interpretability.

Steps:

1. Retrieve relevant clinical guidelines
2. Compare model outputs with standards
3. Generate a clinical summary

Knowledge sources include:

- APA guidelines
- WHO mental health standards

Purpose:
Ensure transparent and explainable AI outputs.

DUAL-MODAL FUSION MODEL

Audio and text features are combined using a:

Gated Dual-Modal Fusion Architecture
Mechanism:

- Sigmoid gating mechanism
- Dynamically weights each modality

Example:

- Poor audio → rely more on text
- Short responses → rely more on speech patterns

Final output:
Depression risk classification

HANDLING CLASS IMBALANCE

Clinical datasets often contain uneven class distribution.

Solution used:

Weighted Cross-Entropy Loss

Benefits:

- Improves detection of minority class
- Maintains screening sensitivity
- Reduces bias toward majority class

DATA SOURCES OVERVIEW

The proposed system consists of three major components:

1. Speech Analysis Pipeline
2. Transcript-Based Analysis
3. RAG-Based Reasoning System

The system uses multiple datasets for training and feature learning:

1. DAIC-WOZ (Primary Clinical Dataset)
2. RAVDESS (Emotion Recognition Dataset)
3. Stress Detection Datasets

These datasets help the model learn depression, emotion, and stress patterns in speech.

Each dataset supports different components of the multimodal pipeline.

DAIC-WOZ DATASET

Distress Analysis Interview Corpus – Wizard of Oz

Source: University of Southern California

Dataset features:

- Clinical interview recordings
- Audio + transcripts
- PHQ-8 depression severity scores

Format:

- 16 kHz WAV audio
- Time-aligned transcripts

Role in project:
Primary dataset used for depression detection training.

RAVDESS DATASET

Ryerson Audio-Visual Database of Emotional Speech and Song

Purpose:
Used for emotion recognition pre-training.

Contains:

- Professional actor recordings
- 8 emotional categories
- High-quality studio audio

Benefit:
Helps model learn basic emotional speech patterns before clinical training.

STRESS DETECTION DATASETS

Stress-related speech datasets are included to help the model understand stress-induced speech variations.

These datasets provide:

- Speech recordings under stress conditions
- Behavioral markers such as hesitation and pitch variation
- Emotional stress indicators

Purpose: Improve model ability to detect mental strain and psychological stress signals.

DATASET PREPARATION PROCESS

The dataset preparation pipeline includes:

- Data source identification
- Data access approval and licensing
- Dataset quality assessment
- Speech-text alignment
- Feature extraction
- Train / validation / test split

Goal:
Prepare a reliable dataset for model training.

EXPLORATORY DATA ANALYSIS (EDA)

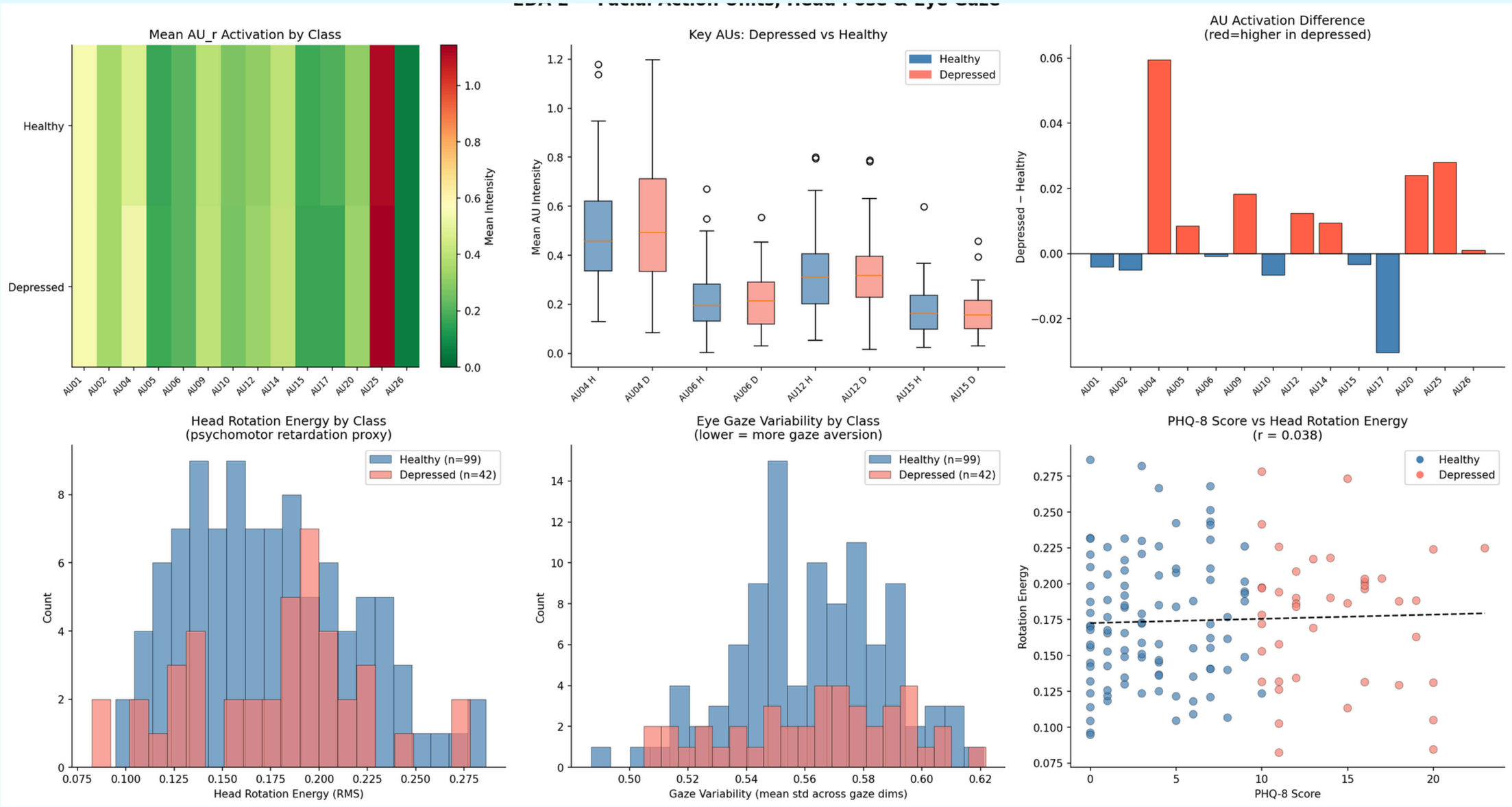
Exploratory Data Analysis was performed to understand dataset characteristics.

EDA tasks included:

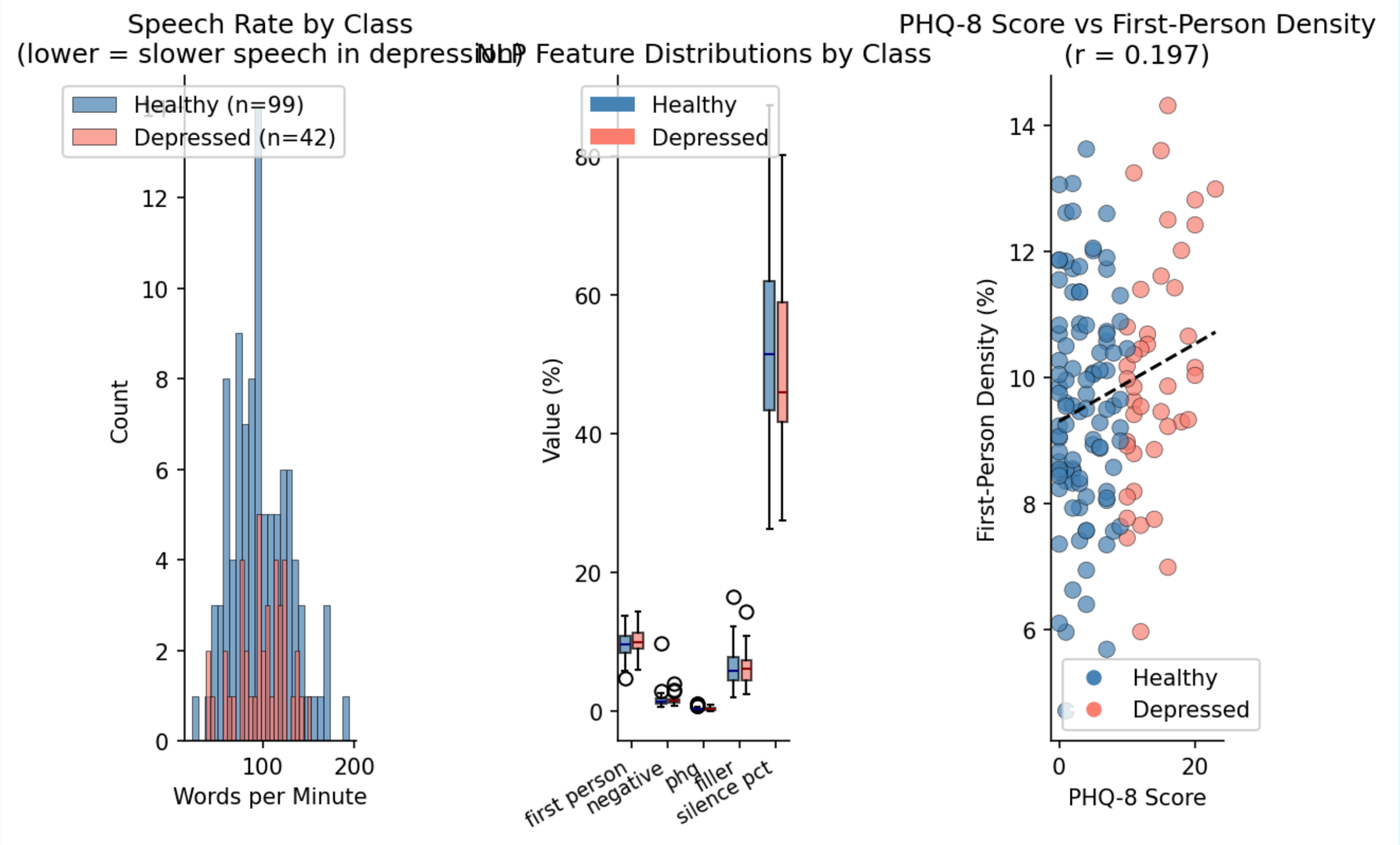
- Distribution of PHQ-8 depression scores
- Audio duration analysis
- Speech feature distribution
- Transcript length analysis
- Class imbalance inspection

EDA helps ensure data quality and model readiness.

FACIAL & GAZE BIOMARKERS OF DEPRESSION



LINGUISTIC MARKERS OF DEPRESSION IN CLINICAL INTERVIEWS



CONCLUSION

This milestone establishes the data foundation for the project.

Key achievements:

- Identified multiple clinical speech datasets
- Designed multimodal data pipeline
- Prepared datasets for feature extraction
- Ensured ethical and research-compliant data usage

Next step:
Model training and multimodal fusion implementation.

CONTRIBUTIONS

Drashti Shah -Dataset Preparation

G Hamsini -Dataset Preparation

Om Aryan - Exploratory Data Analysis (EDA)

Pankaj Sahu - Exploratory Data Analysis (EDA)

Mahi Mugdal - Documentation

THANK YOU